



Re: github

From Vaestermark, Aake <avaeste@okstate.edu>

Date Thu 1/8/2026 9:34 AM

To Capik, Sarah <scapik@okstate.edu>; Ussery, Dave <david.ussery@okstate.edu>

Hi Sarah

Here is the instruction for ios

The main difference is: xcode (not rtools)

See if you can follow these instructions and email me if you need help

Best regards

Aake

Instruction for iOS

For Apple silicon (M1,2,..) Macs:

[R-4.5.2-arm64.pkg](#)

<https://cran.r-project.org>

Download R

What I have: Mac OS ventura 13.5, M1 chip (not intel)

<https://posit.co/download/rstudio-desktop/>

Download R studio

Drag **RStudio** into the **Applications** folder.

###

Rosetta is not installed - does it matter for R bio tools? No...

same as before:

```
install.packages('RCurl')
```

```
install.packages("rmarkdown")
```

```
install.packages("ape")
```

```
install.packages("data.table")
```

```
install.packages("fmsb")
```

```
install.packages("ggplot2")

install.packages("gplots")

install.packages("grImport")

install.packages("gridExtra")

install.packages("pheatmap")

install.packages("rentrez")

install.packages("seqinr")

install.packages("BiocManager", repos = "http://cran.us.r-project.org")
library(BiocManager)
BiocManager::install("msa", force = TRUE)
BiocManager::install("Biostrings", force = TRUE)
library(msa) # hit n for update none
library(Biostrings) # hit n for update none

###

setwd("/Users/aakevaestermark/Downloads")
```

From this GitHub repository, download:

- RBiotoools_0.5.6.tar.gz
Click the file, then **Download button (not Raw)**, and save it in your working directory.

```
###
install.packages("RBiotoools_0.5.6.tar.gz", repos = NULL, type = "source", quiet = TRUE, verbose = FALSE)
install.packages("RBiotoools_0.5.6.tar.gz", repos = NULL, type = "source" )
```

ERROR: dependency 'stringr' is not available for package 'Rbiotoools'
Perhaps try a variation of:
install.packages('stringr')

Command + blank space: term....

Do you need: xcode or rosetta? (Not Rosetta- at least not on this MacBook Air)

xcode-select --install

Verify, then restart R

xcode-select -p

```
install.packages("RBiotoools_0.5.6.tar.gz", repos = NULL, type = "source" )
###
```

```
library(RBiotoools)
eColi <- c("AP012306", "U00096", "CP000802", "CP000800", "AP009378",
"FM180568", "CU928163", "CP008957", "CP027027", "CP026802", "CP026877", "CP026793", "CP015831")
```

invisible(suppressWarnings(suppressMessages(dendrogram16S(eColi)))) # downloading

###

createAtlas(eColi[1]) # it is saved in your working dir
plotUsage(eColi[1]) # hit zoom button
plotHeatMapCodon(eColi) # it is saved in your working dir
#pwalgn fix - blast matrix exercise

###

BiocManager::install("pwalgn")
library(pwalgn)
myRunLinclust <- runLinclust
myRunLinclust <- edit(myRunLinclust)
#on line 133, add this prefix pwalgn::pairwiseAlignment to call, then save.
#On line 141: prefix pid call with: pwalgn::pid()
proteinGrouping <- myRunLinclust(eColi)
plotBlastMatrix(proteinGrouping)

###

Aake

From: Vaestermark, Aake <avaeste@okstate.edu>

Sent: Wednesday, January 7, 2026 9:51 AM

To: Capik, Sarah <scapik@okstate.edu>; Ussery, Dave <david.ussery@okstate.edu>

Subject: Fw: github

Hi Sarah

You are welcome to send any questions directly to me.

you are right -- my instructions assume Windows.

On mac there is no rtools, you need to install xcode: xcode-select --install

Thanks for spotting this.

Aake

From: Ussery, Dave <david.ussery@okstate.edu>

Sent: Wednesday, January 7, 2026 8:55 AM

To: Vaestermark, Aake <avaeste@okstate.edu>

Subject: Fwd: github

Hi Aake,

Sarah is a new assistant professor at the Vet School, who wants to take my class. Can you have a look at her email?

Thanks! 🙏

Will be in my office in roughly 30 minutes

- Dave